

Independent Study Report: Exploring argumentative structures for domain adaptation in conversation prediction tasks

Karthik Sairam

University of Colorado Boulder
karthik.sairam@colorado.edu

Abstract

This project investigates methods for identifying and representing argumentative structures in conversational data, aiming to inform future approaches for enhancing neural conversation models. Focusing on the r/ChangeMyView corpus, this research pursued several key avenues. I first conducted a detailed manual annotation effort on over 500 utterance pairs, classifying reply-to-parent relations as 'Attack', 'Support', or 'Neutral'. This revealed a prevalence of 'Attack' relations and distinct transition patterns in the argumentative flow. I then converted the annotated data into a directed graph representation to capture the structural relationships within conversations. Additionally, I performed initial experiments using a Large Language Model (Llama-3.2-1b) with simple prompting for binary (Attack/Support) and ternary (Attack/Support/Neutral) classification. The LLM experiments showed significant discrepancies compared to human annotations, particularly a strong bias towards predicting 'Support' in the binary task and 'Neutral' in the ternary task. These results underscore the challenges of zero-shot classification for nuanced argumentative relations and highlight the critical role of high-quality, structured datasets in understanding and modeling conversational argumentation for domain adaptation.

1 Introduction

Understanding how participants make claims, provide evidence, and respond to opposing viewpoints at the utterance level is essential for building dialogue systems that can truly grasp and engage in complex conversations. Identifying how arguments unfold – how ideas are supported, challenged, or met with neutral responses – is key to developing more intelligent and nuanced conversational AI. This research project delves into the nature of these argumentative structures within conversational data, specifically focusing on identifying utterance-level relationships like support, attack,

and neutral stances. The goal is to uncover patterns and representations that could potentially be integrated with neural models. The approach taken was to investigate methods for analyzing this symbolic layer, including detailed manual annotation of conversational data, representing these structures as directed graphs, and exploring the capabilities of Large Language Models (LLMs) through prompting for automated classification. This work emphasizes that effectively leveraging argumentative structures requires a deep empirical understanding of the data itself, which was a central focus of the efforts of this research.

2 Related Work

Research into computationally identifying and utilizing argumentative elements in text has explored various approaches, from analyzing components within a single text to capturing relationships across turns in a discussion, often with the goal of improving downstream NLP tasks.

Relevant to this work is the research by Bagyasree Sudharsan, which investigated using domain-agnostic indicators to score persuasiveness in varied texts, including the CMV dataset. This prior work focused on segmenting text and identifying argumentation components (like premises and claims) and their associated semantic types (such as ethos, logos, and pathos), using these as symbolic inputs to enhance transformer-based models for predicting persuasiveness.

Another related study (Öcal et al., 2021) analyzed reasoning behavior specifically within the original posts (OPs) of Reddit's Change My View submissions, annotating claims, rationales, and types of supporting statements such as personal experience, domain expertise, external sources, and definitions to understand how users justify their views. More recently, (Alsinet et al., 2021) explored the use of Graph Neural Networks (GNNs) for argumentation reasoning on Reddit conversa-

tions, representing debates as Debate Trees and Bipartite Debate Graphs to compute measures like polarization. Furthermore, (Fabbri et al., 2021) introduced the ConvoSumm benchmark for conversation summarization, including Reddit data, and demonstrated that incorporating argument mining through graph construction can improve abstractive summarization performance.

Building upon the idea of leveraging explicit argumentative information from the CMV corpus, this research shifts focus from intra-utterance components and semantic types to the inter-utterance relationships (Attack, Support, Neutral) that define the structural flow of argumentation within conversation threads, with the aim of informing conversation prediction tasks.

3 Data

A major aspect of this project involved the careful selection and analysis of conversational datasets that exhibit clear argumentative dynamics. Data from the r/ChangeMyView (CMV) subreddit was utilized and accessed via the Convokit corpus. The CMV forum is particularly suitable for studying argumentation as it explicitly encourages users (from Reddit, a social media platform) to present a view and engage in discussions where others attempt to change it, leading to rich argumentative exchanges. Understanding the complexity of these dialogues necessitated a focused effort on qualitative analysis of data and, crucially, detailed manual annotation. This process was undertaken to explicitly mark and categorize the relationships between these utterance pairs, providing the foundational empirical basis for exploring potential structural representations. To ensure the quality and consistency of this annotation, an inter-annotator agreement study was conducted early in the process.

An 'utterance' in the CMV data can be an original comment or a reply to another comment. Another important feature of this dataset is the existence of a variable that specifies if the OP's (Original Poster) mind was changed through the replies and comments on their post.

4 Methods

4.1 Manual Annotation

The methodology for manual annotation centered on a detailed analysis of the CMV data to identify and characterize its inherent argumentative structures at the utterance level. This involved manually

annotating over 500 pairs of utterances, classifying the relationship between a comment and its direct parent. Based on the nature of argumentative interactions observed in the CMV data and building upon existing work in argumentation analysis, I developed a specific annotation scheme classifying the relationship between an utterance and its parent into one of three categories: Attack, Support, or Neutral.

To refine this scheme and ensure annotation consistency, an initial pilot annotation study was conducted by two annotators (a collaborator and myself) on a subset of 120 utterance pairs. Using the preliminary guidelines, the annotators classified the relationship for each pair. Upon comparing annotations, disagreements were observed in approximately 30 of these 120 instances. Analysis of these disagreements revealed ambiguities in the initial guidelines, particularly concerning subtle forms of attack or support, and the nuances of neutral exchanges like follow-up questions.

Hence, detailed annotation guidelines were developed to ensure consistency and clarity during this process. The full annotation guidelines are presented in [A](#).

The Attack category was used when an utterance directly contradicted, questioned the validity of, or presented counter-arguments against its parent comment, including instances of disagreement that might be softened by polite language. The Support category was applied when an utterance provided evidence, reasoning, or agreement in favor of the parent comment. The Neutral category captured relationships that were not directly argumentative, such as follow-up questions, requests for clarification, irrelevant remarks, or acknowledgments of a view change (Delta awarded comments). Special attention was given to edge cases, such as distinguishing between genuine knowledge-seeking questions and rhetorical questions that imply disagreement, and handling agreement followed by a counter-point, which was treated as an Attack based on common conversational structures.

The complete code for helping with the manual annotation can be found in [this repository](#).

4.2 Graph Construction

To represent the annotated argumentative structures in a format suitable for further analysis and potential computational modeling, the annotated utterance pairs were converted into a directed graph.

This approach allows for the visualization and analysis of the overall conversational structure, capturing not just the direct parent-child relationships but also the type of argumentative interaction occurring along those links.

The full code can be found in [this GitHub repository](#). A Python script utilizing the networkx library was developed for this purpose. The script reads the original conversation data and the manual annotations. It identifies all utterances that were either directly annotated or are the parent of an annotated utterance, creating nodes for each of these relevant turns. Directed edges are then added between parent and child utterances, mirroring the conversational reply structure. Crucially, the 'Attack', 'Support', or 'Neutral' labels from the manual annotations are assigned as attributes to the corresponding edges in the graph. Additional metadata from the original utterances, such as user ID, text content, and timestamp, were included as node attributes to enrich the graph representation. The resulting graph provides a structured representation of the annotated conversational segments, explicitly encoding the identified argumentative relations. One of the annotated conversations is shown in its graphical form in Figure 1.

4.3 LLM Prompting

Initial Experiment As a complementary approach to the manual annotation and graph-based representation, I explored the capability of a Large Language Model (LLM) to classify the argumentative relationship between utterance pairs. This served as an initial computational baseline to assess how well pre-trained models, with minimal fine-tuning, could perform this task based solely on prompt-based inference.

For this initial experiment, I utilized the Llama-3.2-1b model via the transformers library from HuggingFace. The task was framed as a binary classification problem, focusing only on distinguishing between 'Attack' and 'Support' relations, omitting the 'Neutral' category in this phase. A simple prompt (detailed in B) was constructed for each utterance pair, providing the text of the original post (OP) and the reply, and asking the model to classify the relationship as either 'support' or 'attack'.

Revised Experiment Building upon the initial binary classification experiment, a second LLM-based approach was conducted to classify utterance relations into the full set of three categories used

in the manual annotation: 'Attack', 'Support', and 'Neutral'. This aimed to evaluate the LLM's ability to distinguish between all three types of argumentative and non-argumentative interactions present in the dataset using a simple prompting strategy.

Similar to the initial experiment, the Llama-3.2-1b model was used. The prompt was modified to explicitly list 'support', 'attack', and 'neutral' as the possible relation types. The model's output for each OP-reply pair was processed to identify the predicted label among the three categories.

5 Results

Annotation The initial pilot annotation study involving two annotators on a subset of approximately 120 utterance pairs revealed a preliminary distribution of labels. As shown in Figure 2, this initial set contained 84 'Attack' annotations, 24 'Neutral' annotations, and 11 'Support' annotations.

Following the refinement of the annotation guidelines based on the pilot study, the full annotation of approximately 500 utterance pairs was completed. Figure 3 presents the distribution of labels across this larger dataset. The 'Attack' category remained the most frequent, with 366 instances. 'Neutral' accounted for 98 annotations, while 'Support' was the least frequent category, with 35 instances. The distributions in both the pilot and full annotation sets show a consistent pattern where 'Attack' labels are significantly more prevalent than both 'Neutral' and 'Support' labels."

To understand the dynamic flow of argumentative interactions, I also analyzed the transitions between consecutive annotated utterance types using a first-order Markov chain model. Figure 4 illustrates the transition probabilities between the Attack, Support, and Neutral states based on the annotated data. Examining the transitions from each state, it was observed that a 'Support' utterance is most likely to be followed by an 'Attack' with a high probability of 0.82, and less likely by a 'Neutral' utterance (0.20). From an 'Attack' state, the most probable transition is to 'Neutral' (0.44), while transitioning to 'Support' is rare (0.02). 'Neutral' utterances are least likely to lead to either 'Attack' (0.07) or 'Support' (0.09), suggesting they often precede states other than direct argumentative engagement or represent conversational turns that don't fit neatly into the other categories.

LLM Prompting The initial LLM-based prompting experiment framed the task as a binary classi-

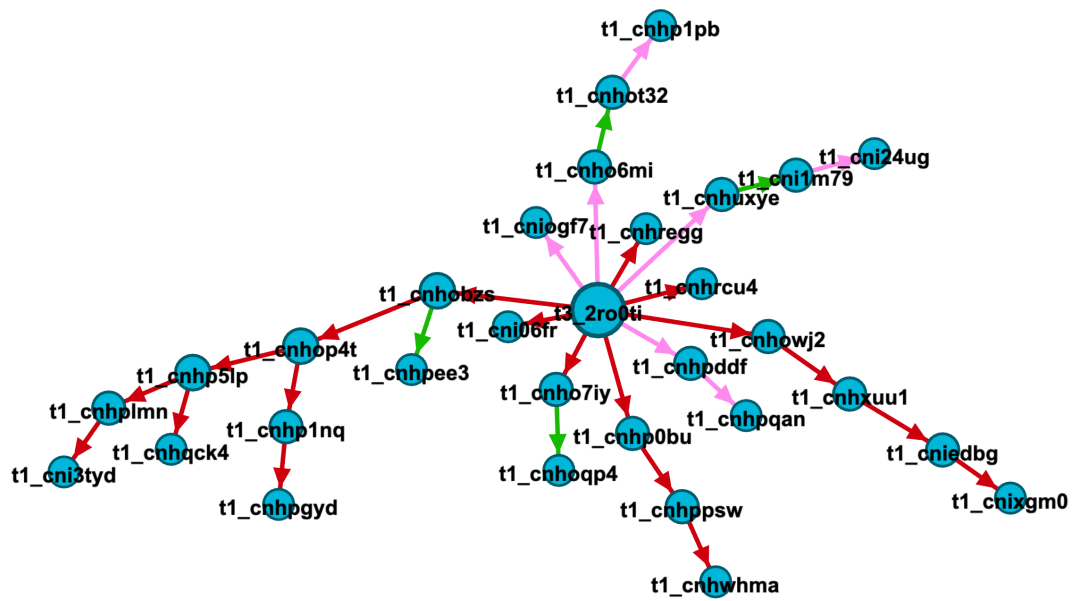


Figure 1: A segment of the constructed directed graph. Nodes represent individual utterances (identified by their IDs). Directed edges connect a reply utterance to its parent. Edge colors indicate the manually annotated argumentative relationship: red for Attack, green for Support, and pink for Neutral.

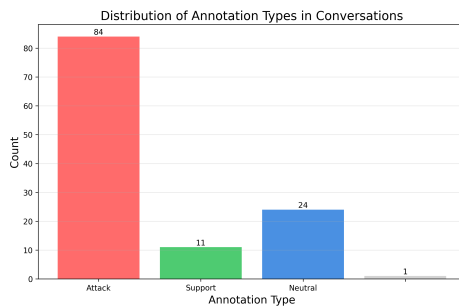


Figure 2: Distribution of preliminary annotation types (Attack, Support, and Neutral) from the initial pilot study on approximately 120 utterance pairs, belonging to one of the collaborators

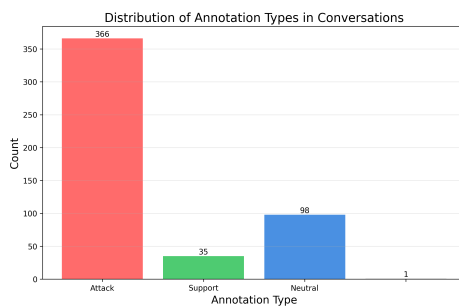


Figure 3: Distribution of preliminary annotation types (Attack, Support, and Neutral) from the full dataset of approximately 500 utterance pairs

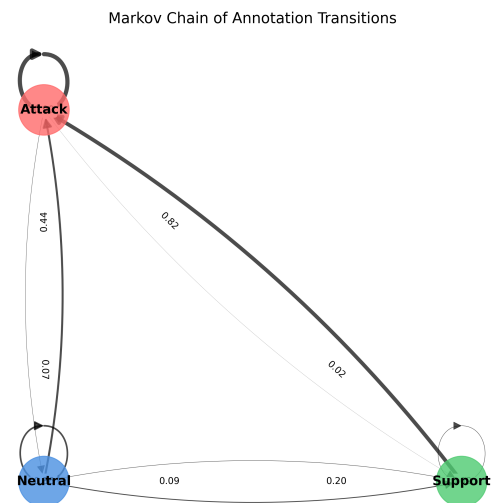


Figure 4: Markov Chain illustrating the transition probabilities between consecutive utterance annotation types

fication problem ('Attack' or 'Support'). The distribution of the model's predictions in this binary setting is presented in Table 1.

The distribution of the LLM's predictions across the three categories for the revised ternary classification is also detailed in Table 1.

6 Discussion

Annotation The observed distributions of annotation types across both the pilot and the full dataset consistently demonstrate a significant imbalance, with 'Attack' annotations being substantially more frequent than 'Support' or 'Neutral'. In the full dataset 3, 'Attack' constitutes the vast majority of annotated relations. This imbalance suggests that contributors are far more likely to engage by challenging existing viewpoints or arguments than by directly supporting them. This characteristic is likely tied to the explicit goal of the subreddit – to change views – which inherently encourages counter-argumentation.

The relative scarcity of 'Support' annotations suggests that direct agreement or reinforcement of a parent comment is less common than disagreement or counter-argumentation in this context. This distributional pattern highlights the importance of models being robust to frequent 'Attack' interactions and suggests that 'Support' and 'Neutral' instances, while less common, may play distinct and crucial roles in steering or concluding argumentative threads. I think that understanding this typical distribution is crucial for developing realistic models of conversational argumentation that reflect the common patterns found in this type of dialogue data, an end goal that I hoped to achieve through this process of qualitative analysis through annotation.

Beyond static counts, the Markov chain analysis of transitions in Figure 4 provides crucial insights into the temporal dynamics of argumentative exchanges in the CMV corpus. The most striking finding is the high probability (0.82) of a 'Support' utterance being followed by an 'Attack'. A possible explanation I have for this counter-intuitive result is that, in this specific argumentative context, offering support to a point might frequently trigger a counter-argument or challenge from another participant, perhaps by strengthening the original view in a way that invites new lines of attack, or by supporting a nuanced point that others disagree with.

Conversely, 'Attack' utterances are relatively likely to be followed by 'Neutral' responses (0.44), which could represent requests for clarification, acknowledgments, or shifts away from direct confrontation. I also don't completely believe this might be fully representative of the complete CMV dataset because a lot of these 'Neutral' label utterances might be irrelevant to the argument.

Graphical Representation The conversion of annotated conversational data into a directed graph provides a powerful representation for analyzing argumentative structure beyond simple pairwise relationships or sequential patterns captured by the Markov chain. This graph explicitly models the network of challenges, supports, and neutral responses within a conversation. This graph representation is a critical step towards integrating symbolic argumentative structure with neural models, potentially serving as an input structure for GNNs or other graph-aware ML techniques.

LLM Prompting The results from the initial LLM-based binary classification experiment present a notable contrast with the distribution observed in our manual annotations. While manual annotation found 'Attack' to be the most prevalent relation in the CMV dataset, the LLM, when forced into a binary 'Support' or 'Attack' choice, predicted 'Support' for the vast majority of utterance pairs (85.96%). This significant discrepancy suggests that a simple prompting approach with a binary classification might not effectively capture the nuances of argumentative interactions in this domain.

Introducing the 'Neutral' category significantly altered the distribution of the LLM's predictions compared to the binary classification task. In the ternary setting, the model exhibited a strong preference for classifying utterance pairs as 'Neutral', accounting for over 82% of predictions, while 'Support' predictions dropped to 17.80% and 'Attack' predictions became extremely rare (0.12%). This, again, contrasts sharply with our manual annotations, where 'Attack' was the most frequent category and 'Neutral' was less common than 'Attack'. The model's high rate of 'Neutral' predictions and near-zero 'Attack' predictions in the ternary setting suggests that, without specific fine-tuning or more complex prompting strategies, the LLM struggles to identify 'Attack' relations and defaults to 'Neutral' or 'Support' for most conversational turns in

Experiment Type	Predicted Relation	Count	Percentage (%)
Binary Classification	Attack	702	14.04
	Support	4298	85.96
Ternary Classification	Attack	6	0.12
	Support	890	17.80
	Neutral	4104	82.08

Table 1: Distribution of predicted argumentative relations from LLM prompting experiments (Binary and Ternary classification) on 5000 utterance pairs from the CMV dataset.

this domain. I think that it’s interesting to note that this prompting experiment underscores the value of the detailed manual annotation effort, and is definitely a key takeaway for me.

7 Conclusion and Future Work

This project explored modeling argumentative structures in conversational data to inform future conversation prediction systems. I conducted detailed manual annotation of utterance relations in the CMV corpus, analyzing the distribution and flow of argumentative acts. This annotated data was then represented as a directed graph to capture conversational structure.

This work provides a foundational dataset and structural representation for conversational argumentation. Future work should focus on leveraging this resource to train or fine-tune computational models, potentially using graph-aware neural networks. Developing more effective LLM strategies for this specific task is also a key direction. The ultimate goal is to integrate these explicit argumentative insights into conversation prediction frameworks to assess their impact on performance.

8 Limitations

A significant limitation of this research was the substantial effort required for the foundational steps of identifying, defining, and annotating argumentative structures within conversational data. Developing a robust annotation scheme suitable for the nuances of dialogue and manually annotating over 500 utterance pairs proved to be a time-intensive undertaking. The inherent subjectivity in classifying argumentative relationships in informal text, as evidenced by the initial inter-annotator disagreements during the pilot study, also presented challenges, even with refined guidelines. Consequently, while the resulting manually annotated dataset offers valuable insights and a crucial foundation, its size is relatively modest for training large-scale

neural models effectively from scratch. The exploration into LLM-based classification was limited to zero-shot prompting experiments. As the results indicated, this simple prompting approach did not effectively capture the nuanced distribution of argumentative relations observed in the human annotations, highlighting that more sophisticated domain adaptation or prompting strategies would likely be necessary for LLMs to perform this task accurately.

9 Acknowledgments

I would like to thank Prof. Maria Pacheco for being an excellent mentor, guiding me through every hurdle I faced and helping me learn a lot of interesting stuff about Neuro-symbolic AI. This research builds upon the original idea of Bagyasree Sudharsan, whose contributions are gratefully acknowledged.

References

- Teresa Alsinet, Josep Argelich, Ramón Béjar, Daniel Gibert, Jordi Planes, and Nil Torrent. 2021. Argumentation reasoning with graph neural networks for reddit conversation analysis. In *Artificial Intelligence Research and Development*, pages 123–131. IOS Press.
- Alexander R. Fabbri, Faiaz Rahman, Imad Rizvi, Borui Wang, Haoran Li, Yashar Mehdad, and Dragomir Radev. 2021. [Convosumm: Conversation summarization benchmark and improved abstractive summarization with argument mining](#). *Preprint*, arXiv:2106.00829.
- Ayşe Öcal, Lu Xiao, and Jaihyun Park. 2021. Reasoning in social media: insights from reddit “change my view” submissions. *Online Information Review*, 45(7):1208–1226.

A Annotation Guidelines

1. Introduction

These guidelines provide instructions for annotating relations between utterances in the CMV

(ChangeMyView) dataset. The goal is to classify the relationship between a given text and its parent comment as either:

- **Attack:** The text disagrees with or challenges the parent comment.
- **Support:** The text agrees with or provides evidence for the parent comment.
- **Neutral:** The text is related to the parent comment but neither attacks nor supports it.

2. General Principles

- **Context is crucial:** Always consider the text in the context of its parent comment.
- **Subjectivity is expected:** Annotation can be subjective. Aim for consistency and refer to examples when in doubt.
- **Softening:** Pay attention to language that softens a disagreement.

3. Annotation Categories

3.1 Attack

- The text directly contradicts the parent comment.
- The text questions the validity of the parent comment's claims.
- The text presents counter-arguments or evidence against the parent comment.
- **Disagree also means attack.**

3.2 Support

- The text provides evidence or reasoning in favor of the parent comment.
- The text agrees with the parent comment's claims.
- The text elaborates on or clarifies the parent comment.

3.3 Neutral

- The text is a follow-up question to the parent comment.
- The text provides additional information without expressing agreement or disagreement.
- The text seeks clarification from the parent comment.

- Delta awarded comments are considered **Neutral**.

- Follow-up questions and answers are **Neutral**, unless they contain additional statements that attack or support.

- When the author changes their view, their reply to their own post is considered **Neutral**.

- Text is irrelevant to the conversation.

4. Specific Cases and Examples

4.1 Follow-up Questions

- Follow-up questions, in isolation, are generally annotated as **Neutral**.

- Example:

- **Parent Comment:** "Almost all mass shootings occur in areas where concealed carry is not permitted."
- **Text:** "Got some stats on this?"
- **Relation:** **Neutral**

4.2 Knowledge-Seeking

- Follow-up questions and answers to those questions are **Neutral** as they are knowledge-seeking, not opinionated.

- Example:

- **Parent Comment:** "And also the harder to defend against them. I think the sweet spot lies somewhere when more people have guns."
- **Text:** "Why?"
- **Relation:** **Neutral**
- **Text:** "Because guns are generally the most viable tool..."
- **Relation:** **Neutral**

4.3 Agreement with Reservations

- A reply that initially agrees with the parent text, and then provides a soft counter-argument is also classified as an **Attack**. The reasoning is that, generally, we as humans tend to disagree with someone in the general structure of 'I see your point, but hear me out <>', or similar forms.

- Example:

- **Parent Comment:** "Mass shootings are incredibly rare in Europe and incredibly common in the US"
- **Text:** "I see your point, but surely you can admit this is a bit of a stretch. 486 people have been killed in the US in the last 13 years. I don't know that I'd call this 'incredibly common.'"
- **Relation:** `Attack`

4.4 Irrelevant Utterances

- Utterances that are irrelevant to the parent comment's topic should be annotated as **Neutral**.
- Example 1:
 - **Parent Comment:** "I am a lawyer... This is one of the most lucid explorations I've seen..."
 - **Text:** "If you're a lawyer, barring the likely tens of thousands of dollars of law school debt, you can probably afford to buy him reddit gold."
 - **Relation:** `Neutral`
- Example 2:
 - **Parent Comment:** "This is your position since you seem to have forgotten: >Cause has to do with agency, choice, and effort"
 - **Text:** Well, you win this round amigo
 - **Relation:** `Neutral`

4.5 Unclear Cases

- If the relationship is unclear, annotate it as the category that seems most fitting.

4.6 Delta Awarded Comments

- Comments that have been awarded a delta (indicating a change in view) should be annotated as **Neutral**.

5. Annotation Procedure

1. Read the parent comment carefully.
2. Read the text to be annotated.
3. Determine the relationship between the text and the parent comment.
4. Select the appropriate category (`Attack`, `Support`, or `Neutral`).
5. Record your annotation.

B LLM Prompt

Task: Determine the relationship between an original post and a reply on the "Change My View" forum. The possible relations are: "support" (agrees with original post) or "attack" (disagrees with or challenges original post).

Original Post: {original_post.text}

Reply: {reply.text}

Relation (support/attack):